

Exploring ∞ -Categories via Simplicial Sets

Zhou Fang

April 21, 2025

Shenzhen Institute for Quantum Science and Engineering,
Southern University of Science and Technology

Part0: Why we need ∞ -category?

An answer from mathoverflow:<https://mathoverflow.net/questions/450835/what-is-the-motivation-for-infinity-category-theory>

- We **discover new phenomena that have no occurrence in classical mathematics**, for instance, there are stable ∞ -categories, i.e., those in which it is amazingly easy to produce a short exact sequence.
- one can **do geometry** in this language, and many moduli spaces get better regularity properties in the higher setting than their truncated counterparts.

An answer from mathoverflow:<https://mathoverflow.net/questions/450835/what-is-the-motivation-for-infinity-category-theory>

- We **discover new phenomena that have no occurrence in classical mathematics**, for instance, there are stable ∞ -categories, i.e., those in which it is amazingly easy to produce a short exact sequence.
- one can **do geometry** in this language, and many moduli spaces get better regularity properties in the higher setting than their truncated counterparts.

PartI: Comprehend simplicial sets from different points of view

I.1:Equivalent definitions of simplicial sets

Definition

The *simplicial category*, denoted by Δ , consists of the following data:

- objects are linearly ordered sets $[n] = \{0 < 1 < 2 < \cdots < n\}$ for $n \geq 0$
- Pick $[m], [n] \in \Delta$, $\text{Hom}_\Delta([m], [n]) = \{\text{nondecreasing map from } [m] \text{ to } [n]\}$

Definition

- (1) Let $\mathcal{C}$ be a category. A simplicial object in $\mathcal{C}$ is a functor $\Delta^{op} \rightarrow \mathcal{C}$.
- (2) A simplicial set is a simplicial object in the category of sets **Set**.
- (3) Let $\mathbf{Set}_\Delta := \text{Fun}(\Delta^{op}, \mathbf{Set})$ be the category of simplicial sets.
- (4) An n -simplex of a simplicial set $S_\bullet$ is an element in $S_\bullet([n]) =: S_n$.
- (5) Define the standard simplex Δ^n as the functor $\text{Hom}_\Delta(-, [n])$

A simplicial set is a family of sets indexed by $n \in \mathbb{N}$ together with morphisms between them.

Definition

The *simplicial category*, denoted by Δ , consists of the following data:

- objects are linearly ordered sets $[n] = \{0 < 1 < 2 < \cdots < n\}$ for $n \geq 0$
- Pick $[m], [n] \in \Delta$, $\text{Hom}_\Delta([m], [n]) = \{\text{nondecreasing map from } [m] \text{ to } [n]\}$

Definition

- (1) Let $\mathcal{C}$ be a category. A *simplicial object* in $\mathcal{C}$ is a functor $\Delta^{op} \rightarrow \mathcal{C}$.
- (2) A *simplicial set* is a simplicial object in the category of sets **Set**.
- (3) Let $\mathbf{Set}_\Delta := \text{Fun}(\Delta^{op}, \mathbf{Set})$ be the category of simplicial sets.
- (4) An *n-simplex* of a simplicial set $S_\bullet$ is an element in $S_\bullet([n]) =: S_n$.
- (5) Define the *standard simplex* Δ^n as the functor $\text{Hom}_\Delta(-, [n])$

A simplicial set is a family of sets indexed by $n \in \mathbb{N}$ together with morphisms between them.

Definition

The simplicial category, denoted by Δ , consists of the following data:

- objects are linearly ordered sets $[n] = \{0 < 1 < 2 < \dots < n\}$ for $n \geq 0$
- Pick $[m], [n] \in \Delta$, $\text{Hom}_{\Delta}([m], [n]) = \{\text{nondecreasing map from } [m] \text{ to } [n]\}$

Definition

- (1) Let $\mathcal{C}$ be a category. A simplicial object in $\mathcal{C}$ is a functor $\Delta^{op} \rightarrow \mathcal{C}$.
- (2) A simplicial set is a simplicial object in the category of sets **Set**.
- (3) Let $\mathbf{Set}_{\Delta} := \text{Fun}(\Delta^{op}, \mathbf{Set})$ be the category of simplicial sets.
- (4) An n -simplex of a simplicial set $S_{\bullet}$ is an element in $S_{\bullet}([n]) =: S_n$.
- (5) Define the standard simplex Δ^n as the functor $\text{Hom}_{\Delta}(-, [n])$

A simplicial set is a family of sets indexed by $n \in \mathbb{N}$ together with morphisms between them.

An n -simplex of simplicial set $S_\bullet$ can be viewed as a morphism (natural transformation from functor Δ^n to functor $S_\bullet$) $\Delta^n \rightarrow S_\bullet$ in **Set** $_\Delta$.

Property

$$S_n \simeq \text{Hom}_{\text{Set}_\Delta}(\Delta^n, S_\bullet)$$

Proof.

By Yoneda lemma, $S_n = S_\bullet([n]) \simeq \text{Hom}_{\text{Set}_\Delta}(\text{Hom}_\Delta(-, [n]), S_\bullet)$. □

Definition

Let $C_\bullet$ be a simplicial object in category $\mathcal{C}$.

Let $\delta_n^i : [n-1] \rightarrow [n]$ denote the strictly increasing map without i in image and let $\sigma_n^i : [n+1] \twoheadrightarrow [n]$ be the nondecreasing function with $\sigma_n^i(i) = \sigma_n^i(i+1)$.

The i -th face operator is the map $d_n^i := C_\bullet(\delta_n^i) : C_n \rightarrow C_{n-1}$. The i -th degeneracy operator is the map $s_i^n := C_\bullet(\sigma_n^i) : C_n \rightarrow C_{n+1}$.

A simplicial object is a functor $\Delta^{op} \rightarrow \mathcal{C}$. The functor is determined by the image of objects and morphisms of Δ^{op} , i.e., a simplicial object is determined by:

- (□) A sequence $\{C_n \in \text{ob}(\mathcal{C})\}_{n \geq 0}$,
together with morphisms $\{\bar{f} : C_n \rightarrow C_m\}_{f \in \text{Hom}_\Delta([m], [n])}$,
satisfying certain compatibility conditions.

This data is huge. However, actually, a simplicial object is determined by a system smaller than (□). Actually, we can only focus on **face and degeneracy operators**, which are just a small part of the system (□).

A simplicial object is a functor $\Delta^{op} \rightarrow \mathcal{C}$. The functor is determined by the image of objects and morphisms of Δ^{op} , i.e., a simplicial object is determined by:

- (□) A sequence $\{C_n \in \text{ob}(\mathcal{C})\}_{n \geq 0}$,
together with morphisms $\{\bar{f} : C_n \rightarrow C_m\}_{f \in \text{Hom}_\Delta([m], [n])}$,
satisfying certain compatibility conditions.

This data is enormous. However, actually, a simplicial object is determined by a system smaller than (□). Actually, we can only focus on **face and degeneracy operators**, which are just a small part of the system (□).

- ($\square'$) A sequence $\{C_n \in \text{ob}(\mathcal{C})\}_{n \geq 0}$,
 together with morphisms $\{d_i^n : C_n \rightarrow C_{n-1}\}_{0 \leq i \leq n, n > 0}$,
 $\{s_i^n : C_n \rightarrow C_{n+1}\}_{0 \leq i \leq n}$, satisfying:
 (i) For $0 \leq i < j \leq n$, we have $d_i^{n-1} \circ d_j^n = d_{j-1}^{n-1} \circ d_i^n$,
 (ii) For $0 \leq i \leq j \leq n$, we have $s_i^{n+1} \circ s_j^n = s_{j+1}^{n+1} \circ s_i^n$,
 (iii) For $0 \leq i \leq n$ and $0 \leq j \leq n+1$, we have

$$d_j^{n+1} \circ s_i^n = \begin{cases} s_{i-1}^{n-1} \circ d_j^n & j < i, \\ \text{id}_{C_n} & j = i \text{ or } j = i+1, \\ s_i^{n-1} \circ d_{j-1}^n & j > i+1. \end{cases}$$

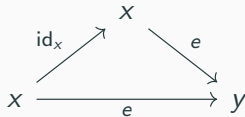
Property

A system $\square'$ determines a unique simplicial object in $\mathcal{C}$ whose face operators and degeneracy operators are $\{d_i^n\}$ and $\{s_i^n\}$ in $\square'$. Conversely, the face operators and degeneracy operators of a simplicial object satisfy conditions (i), (ii), and (iii) in $\square'$.

I.2: Dimension and skeleton of simplicial sets

Definition

Let $S_\bullet$ be a simplicial set. An n -simplex σ is called degenerate if $\sigma \in \text{Im}(s_i^{n-1})$, where s_i^{n-1} is the degeneracy operator.



Example

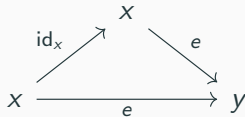
For each $x \in S_0$, denote $s_0^0(x) =: \text{id}_x \in S_1$. ($d_0^1 s_0^0(x) = d_1^1 s_0^0(x) = x$)

Example

Let $e : x \rightarrow y$ is a 1-simplex of simplicial set $S_\bullet$. (e.g. $d_2^2(s_0^1(e)) = s_0^0 d_1^1(e) = s_0^0(x) = \text{id}_x$)

Definition

Let $S_\bullet$ be a simplicial set. An n -simplex σ is called degenerate if $\sigma \in \text{Im}(s_i^{n-1})$, where s_i^{n-1} is the degeneracy operator.



Example

For each $x \in S_0$, denote $s_0^0(x) =: \text{id}_x \in S_1$. ($d_0^1 s_0^0(x) = d_1^1 s_0^0(x) = x$)

Example

Let $e : x \rightarrow y$ is a 1-simplex of simplicial set $S_\bullet$. (e.g. $d_2^2(s_0^1(e)) = s_0^0 d_1^1(e) = s_0^0(x) = \text{id}_x$)

Definition

Let $S_\bullet$ be a simplicial set and $k \in \mathbb{Z}$. We say $S_\bullet$ has dimension $\leq k$ if every n -simplex of $S_\bullet$ is degenerate for $n > k$. Let $k \geq 0$. We say $S_\bullet$ has dimension k if $S_\bullet$ has dimension $\leq k$ but does not have dimension $\leq k - 1$.

Example

Δ^n has dimension n .

To compute the dimension of a simplicial set, it is always useful to know whether an n -simplex is degenerate.

Property

Let $\tau : \Delta^n \rightarrow S_\bullet$ be an n -simplex of simplicial set $S_\bullet$. The following are equivalent:

- (i) τ is degenerate.
- (ii) The map τ factors as a composition $\Delta^n \rightarrow \Delta^m \rightarrow S_\bullet$, where $m < n$
- (iii) The map τ factors as a composition $\Delta^n \xrightarrow{\tau'} \Delta^m \rightarrow S_\bullet$, where τ' is not injective on vertices.

Remark

The i -th vertex $\{i\}$ is a subsimplicial set of Δ^n with $\{i\}[m] = \{\alpha : [m] \rightarrow \{i\} \hookrightarrow [n]\}$

Definition

Let $S_\bullet$ be a simplicial set. We define the subsimplicial set $sk_k(S_\bullet)$ with $sk_k(S_\bullet)_n$ consisting of $\sigma \in S_n$ admitting a factorization $\Delta^n \rightarrow \Delta^m \xrightarrow{\tau} S_\bullet$ for some $m \leq k$. The face and degeneracy operators of $sk_k(S_\bullet)$ are the same as $S_\bullet$. We refer to $sk_k(S_\bullet)$ as the k -skeleton of $S_\bullet$.

- The k -skeleton is the largest simplicial subset of $S_\bullet$ of dimension $\leq k$ (Proved by the property next page)
- If $m \leq n$, then $sk_m \subseteq sk_n$
- $sk_k(S_\bullet)_n = S_n$ for any $n \leq k$ (Any $\sigma : \Delta^n \rightarrow S_\bullet$ can factor as $\Delta^n \xrightarrow{\text{id}} \Delta^n \xrightarrow{\sigma} S_\bullet$. Since $n \leq k$, by definition $\sigma \in sk_n(S_\bullet)$.)
- $S_\bullet = \bigcup_k sk_k(S_\bullet)$ ($sk_k(S_\bullet)_n = S_n$ for any $n \leq k$)
- Let σ be nondegenerate n -simplex of $S_\bullet$. Then $\sigma \in sk_k(S)_n$ if and only if $n \leq k$ (If $n \leq k$, $\sigma \in sk_k(S_\bullet)_n$. If σ contained in $sk_k(S_\bullet)_n$, then σ can factor as $\Delta^n \rightarrow \Delta^m \rightarrow S$ for some $m \leq k$. If $m < n$, then σ degenerates, leading to a contradiction. So $m \geq n$. Since $m \leq k$, we have $n \leq k$.)

- The k -skeleton is the largest simplicial subset of $S_\bullet$ of dimension $\leq k$ (Proved by the property next page)
- If $m \leq n$, then $sk_m \subseteq sk_n$
- $sk_k(S_\bullet)_n = S_n$ for any $n \leq k$ (Any $\sigma : \Delta^n \rightarrow S_\bullet$ can factor as $\Delta^n \xrightarrow{\text{id}} \Delta^n \xrightarrow{\sigma} S_\bullet$. Since $n \leq k$, by definition $\sigma \in sk_n(S_\bullet)$.)
- $S_\bullet = \bigcup_k sk_k(S_\bullet)$ ($sk_k(S_\bullet)_n = S_n$ for any $n \leq k$)
- Let σ be nondegenerate n -simplex of $S_\bullet$. Then $\sigma \in sk_k(S)_n$ if and only if $n \leq k$ (If $n \leq k$, $\sigma \in sk_k(S_\bullet)_n$. If σ contained in $sk_k(S_\bullet)_n$, then σ can factor as $\Delta^n \rightarrow \Delta^m \rightarrow S$ for some $m \leq k$. If $m < n$, then σ degenerates, leading to a contradiction. So $m \geq n$. Since $m \leq k$, we have $n \leq k$.)

- The k -skeleton is the largest simplicial subset of $S_\bullet$ of dimension $\leq k$ (Proved by the property next page)
- If $m \leq n$, then $sk_m \subseteq sk_n$
- $sk_k(S_\bullet)_n = S_n$ for any $n \leq k$ (Any $\sigma : \Delta^n \rightarrow S_\bullet$ can factor as $\Delta^n \xrightarrow{\text{id}} \Delta^n \xrightarrow{\sigma} S_\bullet$. Since $n \leq k$, by definition $\sigma \in sk_n(S_\bullet)$.)
- $S_\bullet = \bigcup_k sk_k(S_\bullet)$ ($sk_k(S_\bullet)_n = S_n$ for any $n \leq k$)
- Let σ be nondegenerate n -simplex of $S_\bullet$. Then $\sigma \in sk_k(S)_n$ if and only if $n \leq k$ (If $n \leq k$, $\sigma \in sk_k(S_\bullet)_n$. If σ contained in $sk_k(S_\bullet)_n$, then σ can factor as $\Delta^n \rightarrow \Delta^m \rightarrow S$ for some $m \leq k$. If $m < n$, then σ degenerates, leading to a contradiction. So $m \geq n$. Since $m \leq k$, we have $n \leq k$.)

- The k -skeleton is the largest simplicial subset of $S_\bullet$ of dimension $\leq k$ (Proved by the property next page)
- If $m \leq n$, then $sk_m \subseteq sk_n$
- $sk_k(S_\bullet)_n = S_n$ for any $n \leq k$ (Any $\sigma : \Delta^n \rightarrow S_\bullet$ can factor as $\Delta^n \xrightarrow{\text{id}} \Delta^n \xrightarrow{\sigma} S_\bullet$. Since $n \leq k$, by definition $\sigma \in sk_n(S_\bullet)$.)
- $S_\bullet = \bigcup_k sk_k(S_\bullet)$ ($sk_k(S_\bullet)_n = S_n$ for any $n \leq k$)
- Let σ be nondegenerate n -simplex of $S_\bullet$. Then $\sigma \in sk_k(S)_n$ if and only if $n \leq k$ (If $n \leq k$, $\sigma \in sk_k(S_\bullet)_n$. If σ contained in $sk_k(S_\bullet)_n$, then σ can factor as $\Delta^n \rightarrow \Delta^m \rightarrow S$ for some $m \leq k$. If $m < n$, then σ degenerates, leading to a contradiction. So $m \geq n$. Since $m \leq k$, we have $n \leq k$.)

- The k -skeleton is the largest simplicial subset of $S_\bullet$ of dimension $\leq k$ (Proved by the property next page)
- If $m \leq n$, then $sk_m \subseteq sk_n$
- $sk_k(S_\bullet)_n = S_n$ for any $n \leq k$ (Any $\sigma : \Delta^n \rightarrow S_\bullet$ can factor as $\Delta^n \xrightarrow{\text{id}} \Delta^n \xrightarrow{\sigma} S_\bullet$. Since $n \leq k$, by definition $\sigma \in sk_n(S_\bullet)$.)
- $S_\bullet = \bigcup_k sk_k(S_\bullet)$ ($sk_k(S_\bullet)_n = S_n$ for any $n \leq k$)
- Let σ be nondegenerate n -simplex of $S_\bullet$. Then $\sigma \in sk_k(S)_n$ if and only if $n \leq k$ (If $n \leq k$, $\sigma \in sk_k(S_\bullet)_n$. If σ contained in $sk_k(S_\bullet)_n$, then σ can factor as $\Delta^n \rightarrow \Delta^m \rightarrow S$ for some $m \leq k$. If $m < n$, then σ degenerates, leading to a contradiction. So $m \geq n$. Since $m \leq k$, we have $n \leq k$.)

Property

Let $S_\bullet$ be a simplicial set and k be an integer. Then for any simplicial set T of dimension $\leq k$, the morphism induced by inclusion $sk_k(S) \hookrightarrow S$ induces a bijection:

$$\mathrm{Hom}_{\mathbf{Set}_\Delta}(T, sk_k(S)) \rightarrow \mathrm{Hom}_{\mathbf{Set}_\Delta}(T, S)$$

For any subsimplicial set T of dimension $\leq k$, the image of T under inclusion $T \subseteq S_\bullet$ is contained in $sk_k(S_\bullet)$. Hence, **the k -skeleton is the largest simplicial subset of $S_\bullet$ of dimension $\leq k$.**

I.3 Build Simplicial sets by Δ^n

Slogan: The standard simplex Δ^n is the building block of a simplicial set.

Since the simplicial set $S_\bullet = \bigcup_k \text{sk}_k(S_\bullet)$, we can “build” simplicial sets by “building” $\text{sk}_k(S_\bullet)$.

Main theorem:

Theorem

Let $S_\bullet$ be a simplicial set and $k \geq 0$. The following is a pushout square:

$$\begin{array}{ccc}
 \coprod_{\sigma \in S_k^{nd}} \partial \Delta^k & \hookrightarrow & \coprod_{\sigma \in S_k^{nd}} \Delta^k \\
 \downarrow & & \downarrow \\
 \text{sk}_{k-1}(S) & \hookrightarrow & \text{sk}_k(S)
 \end{array}$$

Definition

We define $\partial\Delta^n$ as the $(n-1)$ -skeleton of Δ^n , which is referred to the boundary of Δ^n .

Remark

$$\begin{aligned}\partial\Delta^n([m]) &= \text{sk}_{n-1}(\Delta^n)_m \\ &= \{\Delta^m \rightarrow \Delta^s \rightarrow \Delta^n \mid s \leq n-1\} \\ &= \{[m] \rightarrow [s] \rightarrow [n] \mid s \leq n-1\} \\ &= \{\alpha \in \text{Hom}_\Delta([m], [n]) \mid \alpha \text{ is not surjective}\}\end{aligned}$$

Construction

Let S_k^{nd} denote the collection of all nondegenerate k -simplices of $S_\bullet$.

Pick any $\sigma \in S_k^{nd}$, σ is a map $\sigma : \Delta^k \rightarrow \text{sk}_k(S_\bullet)$. Since $\partial\Delta^k = \text{sk}_{k-1}(\Delta^k)$, $\partial\Delta^k$ has dimension $\leq k-1$. So

$$\text{Im}(\sigma|_{\partial\Delta^k}) \subseteq \text{sk}_{k-1}(S_\bullet).$$

This defines the map.

$$\coprod_{\sigma \in S_k^{nd}} \partial\Delta^k \longrightarrow \text{sk}_{k-1}(S_\bullet).$$

Theorem

Let $S_\bullet$ be a simplicial set and $k \geq 0$. The following is a pushout square:

$$\begin{array}{ccc} \coprod_{\sigma \in S_k^{nd}} \partial \Delta^k & \hookrightarrow & \coprod_{\sigma \in S_k^{nd}} \Delta^k \\ \downarrow & & \downarrow \\ \mathrm{sk}_{k-1}(S) & \hookrightarrow & \mathrm{sk}_k(S) \end{array}$$

Definition

Let $S_\bullet$ be a simplicial set. The category of simplices of $S_\bullet$, denoted by Δ_S , consists of the following data:

- Objects are pairs $([n], \sigma)$, where $[n] \in \Delta$ and $\sigma \in S_n$
- Pick $([n], \sigma), ([m], \tau)$ in the objects, let

$$\text{Hom}_{\Delta_S}([n], \sigma), ([m], \tau) := \{f \in \text{Hom}_\Delta([n], [m]) \mid \bar{f} : S_m \rightarrow S_n \text{ such that } \bar{f}(\tau) = \sigma\}$$

Let k be an integer, we let $\Delta_{S, \leq k}$ be the full subcategory of Δ_S spanned by those objects $([n], \sigma)$ with $n \leq k$.

Property

Let k be an integer, and $S_\bullet$ be a simplicial set. Then $\text{sk}_k(S)$ is the image of the monomorphism

$$\lim_{\substack{\longrightarrow \\ ([n], \sigma) \in \Delta_{S, \leq k}}} \Delta^n \longrightarrow S$$

Special case: The simplicial set $S_\bullet$ has finite dimension. Then $S_\bullet = \mathrm{sk}_k(S_\bullet)$ for some $k \in \mathbb{N}$ (because $\mathrm{sk}_k(S)$ is the largest subsimplicial set of dimension $\leq k$).

Corollary

Let $S_\bullet$ be a simplicial set with dimension $\leq k$, where k is an integer. Then, the tautological map

$$\lim_{\substack{\longrightarrow \\ ([n], \sigma) \in \Delta_{S, \leq k}}} \Delta^n \longrightarrow S$$

is an isomorphism of simplicial sets.

Applications:

- (Lemma 1.2.3.13) Roughly speaking, when some colimits exist in a full subcategory $\mathcal{U}$ of $\mathbf{Set}_\Delta$, then $\mathcal{U} = \mathbf{Set}_\Delta$. The proof is based on the above process of building $\mathrm{sk}_k(S_\bullet)$ by Δ^n through colimits.
- Construct a simplicial set from a directed graph. (later)

Applications:

- (Lemma 1.2.3.13) Roughly speaking, when some colimits exist in a full subcategory $\mathcal{U}$ of $\mathbf{Set}_\Delta$, then $\mathcal{U} = \mathbf{Set}_\Delta$. The proof is based on the above process of building $\mathrm{sk}_k(S_\bullet)$ by Δ^n through colimits.
- Construct a simplicial set from a directed graph. (later)

Summary:

- A simplicial set is a functor $\mathbf{\Delta}^{op} \rightarrow \mathbf{Set}$. A simplicial set can be determined by a family of sets indexed by $n \in \mathbb{N}$ together with face operators and degeneracy operators.
- The standard simplex Δ^n is the building block of a simplicial set.

Part II: Simplicial sets and other mathematical objects

II.1: Simplicial sets are generalizations of sets and directed graphs

Slogan: A simplicial set with dimension ≤ 0 is a set

Construction

Let $C \in \mathcal{C}$. We can associate a simplicial object $\underline{C}$ as :

$\underline{C}_n = C$ for $n \geq 0$ together with face and degeneracy operators be identities.

Construction

Let $\mathbf{Set}_{\Delta}^{\leq 0}$ denote the category of simplicial sets of dimension ≤ 0 .

- Define the evaluation functor $ev_0 : \mathbf{Set}_{\Delta}^{\leq 0} \rightarrow \mathbf{Set}$, $S_{\bullet} \mapsto S_0$
- Define the constant functor $C_{st} : \mathbf{Set} \rightarrow \mathbf{Set}_{\Delta}^{\leq 0}$

Property

ev_0 and C_{st} form an equivalence of categories between $\mathbf{Set}$ and $\mathbf{Set}_{\Delta}^{\leq 0}$.

Slogan: A simplicial set with dimension ≤ 0 is a set.

Construction

Let $C \in \mathcal{C}$. We can associate a simplicial object $\underline{C}$ as :

$\underline{C}_n = C$ for $n \geq 0$ together with face and degeneracy operators be identities.

Construction

Let $\mathbf{Set}_{\Delta}^{\leq 0}$ denote the category of simplicial sets of dimension ≤ 0 .

- Define the evaluation functor $ev_0 : \mathbf{Set}_{\Delta}^{\leq 0} \rightarrow \mathbf{Set}$, $S_{\bullet} \mapsto S_0$
- Define the constant functor $C_{st} : \mathbf{Set} \rightarrow \mathbf{Set}_{\Delta}^{\leq 0}$

Property

ev_0 and C_{st} form an equivalence of categories between $\mathbf{Set}$ and $\mathbf{Set}_{\Delta}^{\leq 0}$.

Slogan: A simplicial set with dimension ≤ 0 is a set

Construction

Let $C \in \mathcal{C}$. We can associate a simplicial object $\underline{C}$ as :

$\underline{C}_n = C$ for $n \geq 0$ together with face and degeneracy operators be identities.

Construction

Let $\mathbf{Set}_{\Delta}^{\leq 0}$ denote the category of simplicial sets of dimension ≤ 0 .

- Define the evaluation functor $ev_0 : \mathbf{Set}_{\Delta}^{\leq 0} \rightarrow \mathbf{Set}, S_{\bullet} \mapsto S_0$
- Define the constant functor $C_{st} : \mathbf{Set} \rightarrow \mathbf{Set}_{\Delta}^{\leq 0}$

Property

ev_0 and C_{st} form an equivalence of categories between $\mathbf{Set}$ and $\mathbf{Set}_{\Delta}^{\leq 0}$.

Slogan: A simplicial set with dimension ≤ 1 is a directed graph.

Denote the category of simplicial sets with dimension ≤ 1 as $\mathbf{Set}_{\Delta}^{\leq 1}$. Denote the category of directed graphs as $\mathbf{Graph}$.

Construction

(From simplicial sets to directed graphs) Let X be a simplicial set. We associate a graph, denoted by $Gr(X)$ as:

- $Vert(Gr(X)) := X_0$
- $Edge(Gr(X)) := \{\text{nondegenerate 1-simplexes of } X\}$
- For every $e \in Edge(Gr(X))$, let the source $s(e) := d_1^1(e)$, and the target $t(e) := d_0^1(e)$

So we can define a functor $Gr : \mathbf{Set}_{\Delta}^{\leq 1} \rightarrow \mathbf{Graph}$, $X \mapsto Gr(X)$

Slogan: A simplicial set with dimension ≤ 1 is a directed graph.

Denote the category of simplicial sets with dimension ≤ 1 as $\mathbf{Set}_{\Delta}^{\leq 1}$. Denote the category of directed graphs as **Graph**.

Construction

(From simplicial sets to directed graphs) Let X be a simplicial set. We associate a graph, denoted by $Gr(X)$ as:

- $Vert(Gr(X)) := X_0$
- $Edge(Gr(X)) := \{\text{nondegenerate 1-simplexes of } X\}$
- For every $e \in Edge(Gr(X))$, let the source $s(e) := d_1^1(e)$, and the target $t(e) := d_0^1(e)$

So we can define a functor $Gr : \mathbf{Set}_{\Delta}^{\leq 1} \rightarrow \mathbf{Graph}$, $X \mapsto Gr(X)$

Construction

(From directed graphs to simplicial sets) Let G be a directed graph. We can associate a simplicial set $G_\bullet$ with dimension ≤ 1 by computing the pushout square:

$$\begin{array}{ccc} \coprod_{e \in \text{Edge}(G)} \partial\Delta^1 & \hookrightarrow & \coprod_{e \in \text{Edge}(G)} \Delta^1 \\ \downarrow & & \downarrow \\ \coprod_{v \in \text{Vert}(G)} \Delta^0 & \hookrightarrow & G_\bullet \end{array}$$

So we can obtain a functor $\theta : \mathbf{Graph} \rightarrow \mathbf{Set}_\Delta^{\leq 1}$, $G \mapsto G_\bullet$.

Property

The functor Gr and θ provides an equivalence between category $\mathbf{Graph}$ and $\mathbf{Set}_\Delta^{\leq 1}$.

II.2: Topological spaces and simplicial sets

Similar to the definition for topological space, we can define connected components of a simplicial set and define its π_0 .

Definition

Let $S_\bullet$ be a simplicial set and $S'_\bullet$ be a simplicial subset of $S_\bullet$.

- We say $S'_\bullet$ is a **summand** of $S_\bullet$ if the simplicial set $S_\bullet$ decomposes as a coproduct $S'_\bullet \amalg S''_\bullet$.
- We say $S_\bullet$ is **connected** if it is nonempty and every summand $S'_\bullet \subset S_\bullet$ is either empty or coincides with $S_\bullet$.
- We say $S'_\bullet$ is a **connected component** if $S'_\bullet$ is a summand of $S_\bullet$ and is connected.
- Let $\pi_0(S_\bullet)$ denote the set of all connected components of $S_\bullet$.

Construction

(Functor $\text{Sing}_\bullet(-)$, singular simplicial set of topological space) Let $|\Delta^n|$ be the topological space $\{(t_0, t_1, \dots, t_n) \mid \sum_{i=0}^n t_i = 1\} \subseteq \mathbb{R}^{n+1}$

Let X be a topological space. We define a simplicial set $\text{Sing}_\bullet(X)$ by:

- $\text{Sing}_n(X) := \text{Hom}_{\text{Top}}(|\Delta^n|, X)$
- For $\alpha \in \text{Hom}_\Delta([m], [n])$, $\text{Sing}_\bullet(\alpha) : \text{Sing}_n \rightarrow \text{Sing}_m$ is given by the precomposition of the continuous map $|\Delta^m| \rightarrow |\Delta^n|$,
 $(t_0, t_1, \dots, t_n) \mapsto (\sum_{i \in \alpha^{-1}(0)} t_i, \sum_{i \in \alpha^{-1}(1)} t_i, \dots, \sum_{i \in \alpha^{-1}(n)} t_i)$

Definition

Let $S_\bullet$ be a simplicial set and Y be a topological space. We'll say a map of simplicial sets $u : S_\bullet \rightarrow \text{Sing}_\bullet(Y)$ exhibits Y as a geometric realization of $S_\bullet$ if for every topological space, the composition map

$$\text{Hom}_{\mathbf{Top}}(Y, X) \longrightarrow \text{Hom}_{\mathbf{Set}_\Delta}(\text{Sing}_\bullet(Y), \text{Sing}_\bullet(X)) \xrightarrow{(-) \circ u} \text{Hom}_{\mathbf{Set}_\Delta}(S_\bullet, \text{Sing}_\bullet(X))$$

is a bijection.

Property

Let $S_\bullet$ be a simplicial set. (1) The geometric realization always exists. (2) The geometric realization determined up to homeomorphism and depends functorially on $S_\bullet$, so we can denote the geometric realization of $S_\bullet$ as $|S_\bullet|$.

(3) $|\bullet| : \mathbf{Set}_\Delta \rightarrow \mathbf{Top}$, $S_\bullet \mapsto |S_\bullet|$ is a functor.

Corollary

The functor $|\bullet|$ is left adjoint to the functor $\text{Sing}_\bullet(-)$.

Definition

Let $S_\bullet$ be a simplicial set and Y be a topological space. We'll say a map of simplicial sets $u : S_\bullet \rightarrow \text{Sing}_\bullet(Y)$ exhibits Y as a geometric realization of $S_\bullet$ if for every topological space, the composition map

$$\text{Hom}_{\mathbf{Top}}(Y, X) \longrightarrow \text{Hom}_{\mathbf{Set}_\Delta}(\text{Sing}_\bullet(Y), \text{Sing}_\bullet(X)) \xrightarrow{(-) \circ u} \text{Hom}_{\mathbf{Set}_\Delta}(S_\bullet, \text{Sing}_\bullet(X))$$

is a bijection.

Property

Let $S_\bullet$ be a simplicial set. (1) The geometric realization always exists. (2) The geometric realization determined up to homeomorphism and depends functorially on $S_\bullet$, so we can denote the geometric realization of $S_\bullet$ as $|S_\bullet|$.

(3) $|\bullet| : \mathbf{Set}_\Delta \rightarrow \mathbf{Top}$, $S_\bullet \mapsto |S_\bullet|$ is a functor.

Corollary

The functor $|\bullet|$ is left adjoint to the functor $\text{Sing}_\bullet(-)$.

Definition

Let $S_\bullet$ be a simplicial set and Y be a topological space. We'll say a map of simplicial sets $u : S_\bullet \rightarrow \text{Sing}_\bullet(Y)$ exhibits Y as a geometric realization of $S_\bullet$ if for every topological space, the composition map

$$\text{Hom}_{\mathbf{Top}}(Y, X) \longrightarrow \text{Hom}_{\mathbf{Set}_\Delta}(\text{Sing}_\bullet(Y), \text{Sing}_\bullet(X)) \xrightarrow{(-) \circ u} \text{Hom}_{\mathbf{Set}_\Delta}(S_\bullet, \text{Sing}_\bullet(X))$$

is a bijection.

Property

Let $S_\bullet$ be a simplicial set. (1) The geometric realization always exists. (2) The geometric realization determined up to homeomorphism and depends functorially on $S_\bullet$, so we can denote the geometric realization of $S_\bullet$ as $|S_\bullet|$.

(3) $|\bullet| : \mathbf{Set}_\Delta \rightarrow \mathbf{Top}$, $S_\bullet \mapsto |S_\bullet|$ is a functor.

Corollary

The functor $|\bullet|$ is left adjoint to the functor $\text{Sing}_\bullet(-)$.

Property

Let X be a topological space, then $\pi_0(\text{Sing}_\bullet(X)) \simeq \pi_0(X)$. (c.f. Remark 1.2.2.5)

Remark

(Warning 1.2.1.9) For a simplicial set $S_\bullet$, there are more ways to compute π_0 .

- $\pi_0(S_\bullet)$ can be identified with a colimit of the diagram $\Delta^{op} \rightarrow \mathbf{Set}$ given by the simplicial set $S_\bullet$.
- $\pi_0(S_\bullet)$ is the set of connected components of the directed graph $Gr(S_\bullet)$
- ...

Here is a parallel property:

Property

Let $S_\bullet$ be a simplicial set. Then $\pi_0(S_\bullet) \simeq \pi_0(|S_\bullet|)$

Remark

There is a generalization of the above property; see Proposition 1.2.3.15. Also, the concrete construction of geometric realization can be found in Remark 1.2.3.16

II.3: Simplicial set is a generalization of category

Definition

View the linear ordered set $[n]$ as a category. Let $N_{\bullet}(\mathcal{C}) := \text{Fun}(-, \mathcal{C}) : \Delta^{op} \rightarrow \mathbf{Set}$ be a simplicial set. We refer to $N_{\bullet}(\mathcal{C})$ as the nerve of $\mathcal{C}$.

Fact

$N_{\bullet} : \mathbf{Cat} \rightarrow \mathbf{Set}_{\Delta}$, $\mathcal{C} \mapsto N_{\bullet}(\mathcal{C})$ is a functor.

Slogan: A category in **Cat** can be identified with $N_{\bullet}(\mathcal{C})$ in **Set** $_{\Delta}$.

Property

- The functor $N_{\bullet} : \mathbf{Cat} \rightarrow \mathbf{Set}_{\Delta}$ is a fully faithful functor.
- Let $S_{\bullet} \in \mathbf{Set}_{\Delta}$. $S_{\bullet} \simeq N_{\bullet}(\mathcal{C})$ for some $\mathcal{C} \in \mathbf{Cat}$ ($S_{\bullet}$ lies in the image of $N_{\bullet}$) if and only if $S_{\bullet}$ satisfies condition $\star'$:
($\star'$) For $n > 0$ and $0 < i < n$, every morphism $\sigma_0: \Delta_i^n \rightarrow S_{\bullet}$ can be extended uniquely to $\sigma: \Delta^n \rightarrow S_{\bullet}$.

Definition

Let $\mathcal{C}$ be a category. We'll say a map of simplicial sets $u : S_\bullet \rightarrow N_\bullet(\mathcal{C})$ exhibits as the homotopy category of $S_\bullet$ if, for every category $\mathcal{D}$, the composite map

$$\mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D}) \xrightarrow{\cong} \mathrm{Hom}_{\mathbf{Set}_\Delta}(N_\bullet(\mathcal{C}), N_\bullet(\mathcal{D})) \xrightarrow{- \circ u} \mathrm{Hom}_{\mathbf{Set}_\Delta}(S_\bullet, N_\bullet(\mathcal{D}))$$

is bijective.

Property

Let $S := S_\bullet$ be a simplicial set. (1) Homotopy category of $S_\bullet$ exists. (2) The homotopy category is unique up to isomorphism and depends functorially on S . We denote the homotopy category of $S_\bullet$ as hS , and thus $h : \mathbf{Set}_\Delta \rightarrow \mathbf{Cat}$ is a functor.

Corollary

The $h : \mathbf{Set}_\Delta \rightarrow \mathbf{Cat}$ is left adjoint to $N_\bullet : \mathbf{Cat} \rightarrow \mathbf{Set}_\Delta$.

Definition

Let $\mathcal{C}$ be a category. We'll say a map of simplicial sets $u : S_\bullet \rightarrow N_\bullet(\mathcal{C})$ exhibits as the homotopy category of $S_\bullet$ if, for every category $\mathcal{D}$, the composite map

$$\mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D}) \xrightarrow{\cong} \mathrm{Hom}_{\mathbf{Set}_\Delta}(N_\bullet(\mathcal{C}), N_\bullet(\mathcal{D})) \xrightarrow{- \circ u} \mathrm{Hom}_{\mathbf{Set}_\Delta}(S_\bullet, N_\bullet(\mathcal{D}))$$

is bijective.

Property

Let $S := S_\bullet$ be a simplicial set. (1) Homotopy category of $S_\bullet$ exists. (2) The homotopy category is unique up to isomorphism and depends functorially on S . We denote the homotopy category of $S_\bullet$ as hS , and thus $h : \mathbf{Set}_\Delta \rightarrow \mathbf{Cat}$ is a functor.

Corollary

The $h : \mathbf{Set}_\Delta \rightarrow \mathbf{Cat}$ is left adjoint to $N_\bullet : \mathbf{Cat} \rightarrow \mathbf{Set}_\Delta$.

Definition

Let $\mathcal{C}$ be a category. We'll say a map of simplicial sets $u : S_\bullet \rightarrow N_\bullet(\mathcal{C})$ exhibits as the homotopy category of $S_\bullet$ if, for every category $\mathcal{D}$, the composite map

$$\mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D}) \xrightarrow{\cong} \mathrm{Hom}_{\mathbf{Set}_\Delta}(N_\bullet(\mathcal{C}), N_\bullet(\mathcal{D})) \xrightarrow{- \circ u} \mathrm{Hom}_{\mathbf{Set}_\Delta}(S_\bullet, N_\bullet(\mathcal{D}))$$

is bijective.

Property

Let $S := S_\bullet$ be a simplicial set. (1) Homotopy category of $S_\bullet$ exists. (2) The homotopy category is unique up to isomorphism and depends functorially on S . We denote the homotopy category of $S_\bullet$ as hS , and thus $h : \mathbf{Set}_\Delta \rightarrow \mathbf{Cat}$ is a functor.

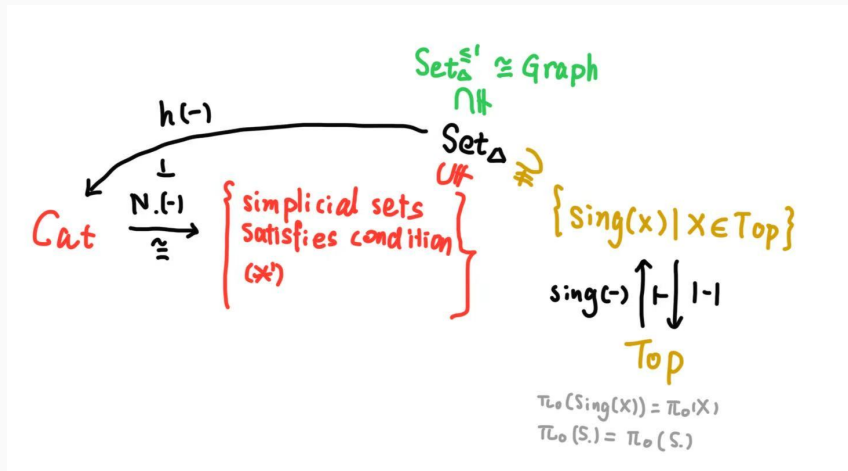
Corollary

The $h : \mathbf{Set}_\Delta \rightarrow \mathbf{Cat}$ is left adjoint to $N_\bullet : \mathbf{Cat} \rightarrow \mathbf{Set}_\Delta$.

Example

$G_\bullet$ is a simplicial set of dimension ≤ 1 and G is the directed graph associated with $G_\bullet$. Then the path category $\text{Path}[G]$ is the homotopy category of $G_\bullet$, see Proposition 1.3.7.5.

Summary:



PartIII: ∞ -category

Definition

We define a sub simplicial set Λ_i^n of Δ^n with
 $(\Lambda_i^n)([m]) := \{\alpha \in \text{Hom}_\Delta([m], [n]) \mid \alpha([m]) \cup \{i\} \neq [n]\}$, where $\{i\}$ is the i -th vertex of Δ^n . We refer to Λ_i^n the i -th horn of Δ^n .

There are three conditions:

- ($\star$) For $n > 0$ and $0 \leq i \leq n$, every morphism of simplicial sets $\sigma_0: \Lambda_i^n \rightarrow S_\bullet$ can be extended to an n -simplex $\sigma: \Delta^n \rightarrow S_\bullet$ (Kan complex)
- ($\star'$) For $n > 0$ and $0 < i < n$, every morphism $\sigma_0: \Lambda_i^n \rightarrow S_\bullet$ can be extended *uniquely* to $\sigma: \Delta^n \rightarrow S_\bullet$.
- ($\star''$) For $n > 0$ and $0 < i < n$, every morphism $\sigma_0: \Lambda_i^n \rightarrow S_\bullet$ can be extended to $\sigma: \Delta^n \rightarrow S_\bullet$ (weak Kan complex)

Remark

(1) “Extend” here means $\sigma|_{\Lambda_i^n} = \sigma_0$ (2) The condition $\star''$ is weaker than $\star$ and $\star'$.

Definition

- A kan complex is a simplicial set satisfying kan extension condition $\star$
- An ∞ -category is a simplicial set satisfying weak kan extension condition $\star''$.

Definition

A functor between ∞ -category is a simplicial map between them.

Remark

(Proposition 1.3.5.2) Let $\mathcal{C}$ be a category. $N_{\bullet}(\mathcal{C})$ is a kan complex if and only if $\mathcal{C}$ is a groupoid.

Definition

- A kan complex is a simplicial set satisfying kan extension condition $\star$
- An ∞ -category is a simplicial set satisfying weak kan extension condition $\star''$.

Definition

A functor between ∞ -category is a simplicial map between them.

Remark

(Proposition 1.3.5.2) Let $\mathcal{C}$ be a category. $N_{\bullet}(\mathcal{C})$ is a kan complex if and only if $\mathcal{C}$ is a groupoid.

Definition

- A kan complex is a simplicial set satisfying kan extension condition $\star$
- An ∞ -category is a simplicial set satisfying weak kan extension condition $\star''$.

Definition

A functor between ∞ -category is a simplicial map between them.

Remark

(Proposition 1.3.5.2) Let $\mathcal{C}$ be a category. $N_{\bullet}(\mathcal{C})$ is a kan complex if and only if $\mathcal{C}$ is a groupoid.

Example

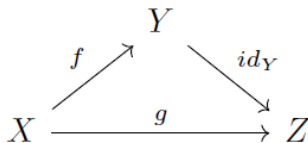
- Any Kan complex is a is an ∞ -category.
- Let X be a topological space. Then $\text{Sing}_\bullet(X)$ is a kan complex and thus an ∞ -category.
- For any category $\mathcal{C}$, $N_\bullet(\mathcal{C})$ satisfying $(\star')$, so $\mathcal{C}$ is an ∞ -cat. (Since $N_\bullet : \mathcal{C} \mapsto N_\bullet(\mathcal{C})$ is fully faithful, we can always identify $\mathcal{C}$ with $N_\bullet(\mathcal{C})$). So, the ordinary category is an ∞ -category.

Definition

Let $\mathcal{C}$ be an ∞ -category. An **object** of $\mathcal{C}$ is an element of $\mathcal{C}_0$. A **morphism** f of $\mathcal{C}$ is an element of $\mathcal{C}_1$. The target of morphism f is $d_0^1(f)$ and the source of f is $d_1^1(f)$. In this case we denote f by $f : d_1^1(f) \rightarrow d_0^1(f)$.

Definition

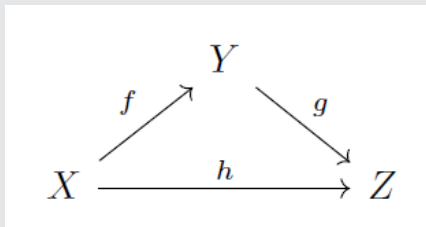
Let $\mathcal{C}$ be an ∞ -category and let $f, g : X \rightarrow Y$ be morphisms in $\mathcal{C}$. A homotopy from f to g is a 2-simplex σ satisfying $d_0^2(\sigma) = id_Y$, $d_1^2(\sigma) = g$, $d_2^2(\sigma) = f$



Homotopy is an equivalence relationship.

Definition

Let $\mathcal{C}$ be an ∞ -category. Let X, Y, Z be objects in $\mathcal{C}$ and $f : X \rightarrow Y, g : Y \rightarrow Z$ be two morphisms of $\mathcal{C}$. We say morphisms $h : X \rightarrow Z$ is the composition of f and g if there exists a 2-simplex with $d_0^2(\sigma) = g, d_1^2(\sigma) = h, d_2^2(\sigma) = f$, denoted by



In this case, we say that σ witnesses h as a composition of f and g .

Useful facts about the composition.

- The composition of f and g exists and **determined up to homotopy**.
($h : X \rightarrow Z$ is a composition of f and g . $h' : X \rightarrow Z$ is a composition of f and g if and only if h' homotopic to h .)
- If f homotopic to f' , g homotopic to g' , h is a composition of f and g , h' is a composition of f' and g' . Then h are homotopic to h' .

Construction

Let $\mathcal{C}$ be an ∞ -cat. We associate an ordinary category $h\mathcal{C}$ with the following data:

- Objects of $h\mathcal{C}$ are objects of $\mathcal{C}$.
- For $X, Y \in h\mathcal{C}$, we define $\text{Hom}_{h\mathcal{C}}(X, Y)$ be the homotopy classes of $\text{Hom}_{\mathcal{C}}(X, Y)$. We refer $[f]$ the homotopy class of $f \in \text{Hom}_{\mathcal{C}}(X, Y)$.

Based on the above facts, the composition is well-defined, i.e., the composition law is unique. For example, $[g][f] = [h]$ with the same notation as above.

Remark

We have two definitions for the homotopy category of an ∞ -category. In fact, they are the same.

Example

The homotopy category $h\mathrm{Sing}_\bullet(X)$ of $\mathrm{Sing}_\bullet(X)$ is the fundamental groupoid $\pi_{\leq 1}(X)$.

“Kerodon: The difference between an ∞ -category S and its homotopy category hS is that the former can contain nontrivial homotopy-theoretic information (encoded by simplices of dimension $n \geq 2$, which can loosely be understood as n -morphisms) which is lost upon passage to the homotopy category hS .”

With composition, we can define isomorphism and invertible morphisms.

Definition

Let $\mathcal{C}$ be an ∞ -category and $f : X \rightarrow Y$ is a morphism in $\mathcal{C}$. f is called an isomorphism if $[f]$ is an isomorphism in $h\mathcal{C}$.

Actually, the property of “morphism invertible” describes Kan complexes.

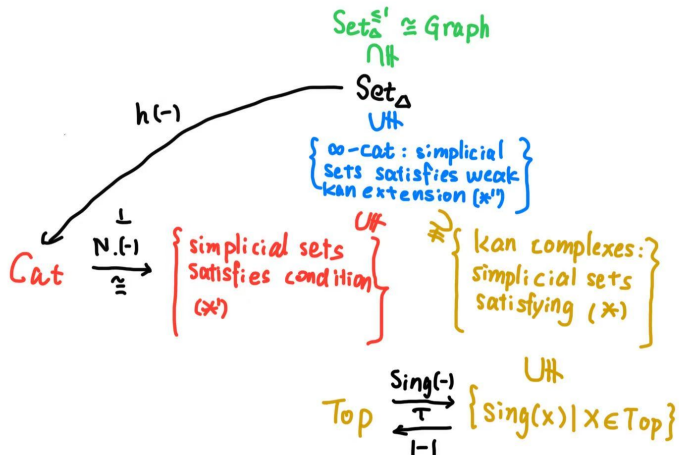
Property

Let $\mathcal{C}$ be an ∞ -category. $\mathcal{C}$ is a Kan complex if and only if any morphisms are invertible.

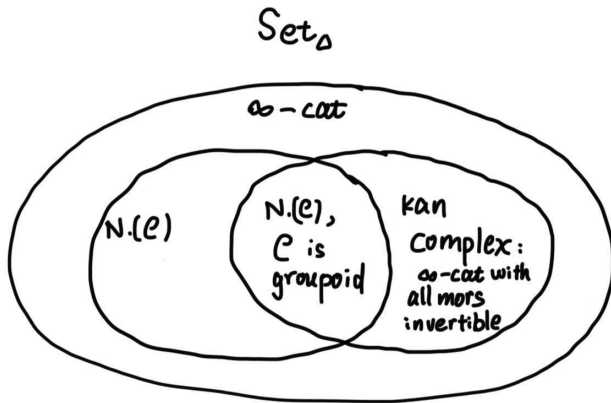
Summary:

- Morphisms in an ∞ -category can be grouped into homotopy classes.
- These classes compose: $[f] \circ [g] = [h]$
- This gives the **homotopy category**.
- $\{\text{categories}\} + \{\text{homotopy theory}\} = \{\infty\text{-categories}\}$

Summary:



Summary:



Reference: Chapter 1 of the Kerodon's online notes (<https://kerodon.net/tag/0000>)

Thanks!